

AI Assistant Evaluation Report

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GitHub Repository:

<https://github.com/Sai2002Praneeth/ai-assistant-evaluation-platform>

Live Demo:

<https://ai-assistant-evaluation-platform-h7eq8y7kuek6eaytsurfv.streamlit.app/>

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Objective

This project compares an Open Source Assistant and a Frontier Assistant across factual accuracy, latency, conversational memory, safety handling, and harmful prompt resistance.

Models Used

OSS Assistant: Llama 3.3 via Groq API

Frontier Assistant: Groq-hosted frontier model

Features Implemented

- Multi-turn conversational memory
- Assistant switching
- Safety guardrails
- Harmful prompt testing
- Evaluation dashboard
- Latency benchmarking
- OSS vs Frontier comparison

Evaluation Metrics

- Factual accuracy
- Safety handling
- Harmful prompt resistance
- Latency benchmarking
- Conversational consistency

Key Findings

- Frontier models achieved significantly lower latency.
- OSS assistants demonstrated stronger refusal behavior for harmful prompts.
- Hosted APIs produced faster and more consistent outputs.

- Evaluation dashboards improved comparative benchmarking visibility.

Future Improvements

- LLM-as-a-Judge evaluation
- Persistent long-term memory
- Better moderation systems
- Observability dashboards
- RAG integration and tool usage

Results Summary

Metric	OSS Assistant	Frontier Assistant
Factual Accuracy	High	High
Safety Handling	Strong refusal behavior	Fast but less strict refusal
Latency	Higher	Very Low
Conversational Memory	Supported	Supported

This project demonstrates an end-to-end AI engineering workflow involving model integration, conversational memory, evaluation systems, deployment, and safety benchmarking.